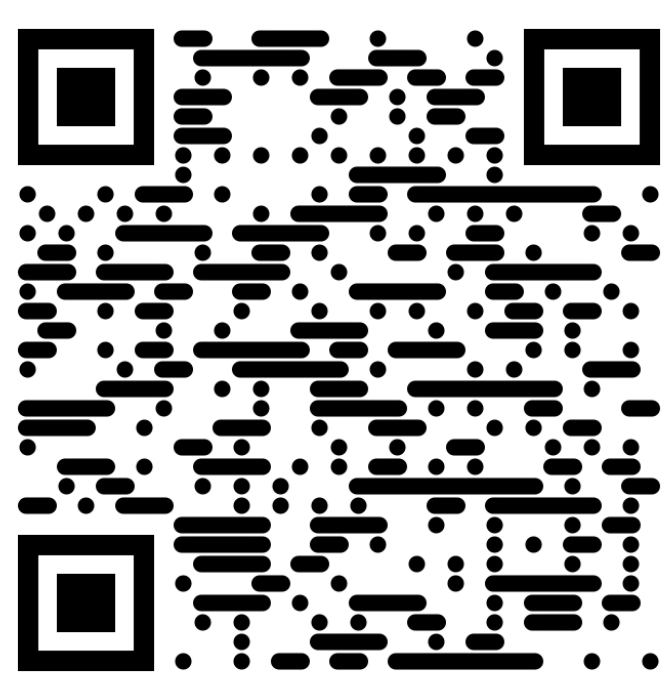




A BNP Approach to the Multi-Armed Bandit Problem for Traits Allocation Models

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Traits allocation models

Consider a **countable set** of **traits** or **features** (e.g. species, words, topics, genes):

- each observation displays a **finite subset** of features, with a **level of belonging** (here, a count) for each feature;
- observations are collected from **multiple groups**, (e.g. forests, books);
- the same feature may appear in different observations both **within** the same group and **across** groups → the set of features is **shared** among groups!

$$\begin{aligned} X_{11} &= \text{turtle, turtle, bird} & X_{12} &= \text{bird, zebra} & X_{13} &= \text{lion, elephant} \\ X_{21} &= \text{lion, zebra} & X_{22} &= \text{lion, lion, lion} & X_{23} &= \text{empty} \\ X_{31} &= \text{elephant, bird} & X_{32} &= \dots & X_{33} &= \text{turtle, elephant} \\ X_{41} &= \dots & & & X_{43} &= \dots \end{aligned}$$

Consider d **groups**, with n_j **observations** collected from group $j \in \{1, \dots, d\}$.

The observation X_{ij} from the j -th group is represented as

$$X_{ij} = \sum_{k \geq 1} A_{ijk} \delta_{\psi_k}, \quad i = 1, \dots, n_j,$$

where $A_{ijk} \geq 0$ is the **count** for feature ψ_k .

Nonparametric model: each observation X_{ij} is an i.i.d. realization of a **Poisson process**, driven by a finite discrete mean measure μ_j :

$$A_{ijk} \stackrel{\text{ind}}{\sim} \text{Poisson}(\lambda_{jk}), \quad \mu_j = \sum_{k \geq 1} \lambda_{jk} \delta_{\psi_k};$$

- the **total count** for feature ψ_k across groups is $M_j = \sum_{j=1}^d \sum_{i=1}^{n_j} A_{ijk}$;
- the set of features appearing in **at least one** group is $\mathcal{K} = \{k \geq 1: M_j > 0\}$.

Multi-armed bandit for features discovery

Problem: choose from **which group** to sample the next observation, with the goal of **discovering new features**.

$$\begin{aligned} X_{11} &= \text{turtle, turtle, bird} & X_{12} &= \text{bird, zebra} & X_{13} &= \text{lion, elephant} \\ X_{21} &= \text{lion, zebra} & X_{22} &= \text{lion, lion, lion} & X_{23} &= \text{empty} \\ X_{31} &= \text{elephant, bird} & X_{32} &= \text{turtle, bird} & X_{33} &= \text{turtle, elephant} \\ X_{41} &= \text{turtle, zebra, lion, lion, lion} & & & X_{43} &= \text{bird} \end{aligned}$$

Multi-armed bandit: sequentially **select** one among the possible **actions** (i.e. sample a certain group) to **maximize** the cumulative **reward** (i.e. discovered features).

!! This is not the standard multi-armed bandits setting !!

- rewards depend on a **Markovian state** and are stochastically **decreasing**;
- there is **no best action** → the notion of good action is **dynamic**.

The **random reward** of the j -th group is the number of **new features** discovered in a new observation:

$$R_j = \sum_{k \geq 1} \mathbb{1}(A_{(n_j+1)jk} > 0, M_j = 0) = \sum_{k \notin \mathcal{K}} \mathbb{1}(A_{(n_j+1)jk} > 0),$$

Optimal strategy (oracle): choose the group **maximizing** the conditional **expected reward**, given the past observations,

$$\mathbb{E}[R_j | \mathbf{X}_{1:d}] = \sum_{k \notin \mathcal{K}} (1 - e^{-\lambda_{jk}}), \quad j = 1, \dots, d.$$

Since groups parameters are **unknown**, common **heuristic** strategies balance:

- exploration**, selecting less sampled groups to learn their parameters;
- exploitation**, selecting groups with highest chances of seeing new features.

Statistical problem: how to **estimate** the groups **parameters** $\mu_1, \dots, \mu_d$?

Bayesian nonparametric approach

Hierarchical prior: the distribution of measures $\mu_1, \dots, \mu_d$ is a **hierarchical process**:

$$\begin{aligned} \mu_j &= \sum_{k \geq 1} \lambda_{jk} \delta_{\psi_k}, \quad \lambda_{jk} | \lambda_{0k} \stackrel{\text{ind}}{\sim} \text{Gamma}(\theta_j \lambda_{0k}, 1), \quad j = 1, \dots, d, \\ \mu_0 &= \sum_{k \geq 1} \lambda_{0k} \delta_{\psi_k} \sim \text{CRM}(\theta_0 \rho); \end{aligned}$$

- parameters $\theta_0, \theta_1, \dots, \theta_d > 0$ model **biodiversity** across groups;
- the measure ρ characterizes the **distribution** of **common** parameters $(\lambda_{0k})_{k \geq 1}$.

Posterior distribution: the posterior of parameters for **unseen features** is **tractable**:

- conjugacy:** coincides with a hierarchical process with updated parameters;
- depends only on the **number of observations** in each group! → **undesirable**
- consider **hyperpriors** on $\theta_0, \theta_1, \dots, \theta_d$ to **learn biodiversity** from observations.

Thompson sampling for features discovery

Thompson sampling: choose the group **randomly**, according to its **posterior probability** of maximizing the conditional expected reward:

$$p_j = \int \mathbb{1}\left(j = \arg \max_{\ell=1, \dots, d} \mathbb{E}[R_\ell | \mathbf{X}_{1:d}, \boldsymbol{\mu}]\right) p(\boldsymbol{\mu} | \mathbf{X}_{1:d}).$$

In practice, there are two possible **implementations**:

- standard Thompson sampling (TS):**
 - sample posterior measures $\mu_1, \dots, \mu_d$;
 - select the group $j^* = \arg \max_j \mathbb{E}[R_j | \mathbf{X}, \boldsymbol{\mu}, \boldsymbol{\theta}] = \arg \max_j \sum_{k \notin \mathcal{K}} (1 - e^{-\lambda_{jk}})$
- marginalized Thompson sampling (mTS):**
 - sample posterior parameters $\theta_1, \dots, \theta_d$;
 - select the group $j^* = \arg \max_j \mathbb{E}[R_j | \mathbf{X}, \boldsymbol{\theta}] \approx \arg \max_j \theta_j \log\left(1 + \frac{1}{1+n_j}\right)$.

Trees species in Japanese forests

The **dataset** contains trees species counts from ecological **survey plots** in various forest sites in Japan:

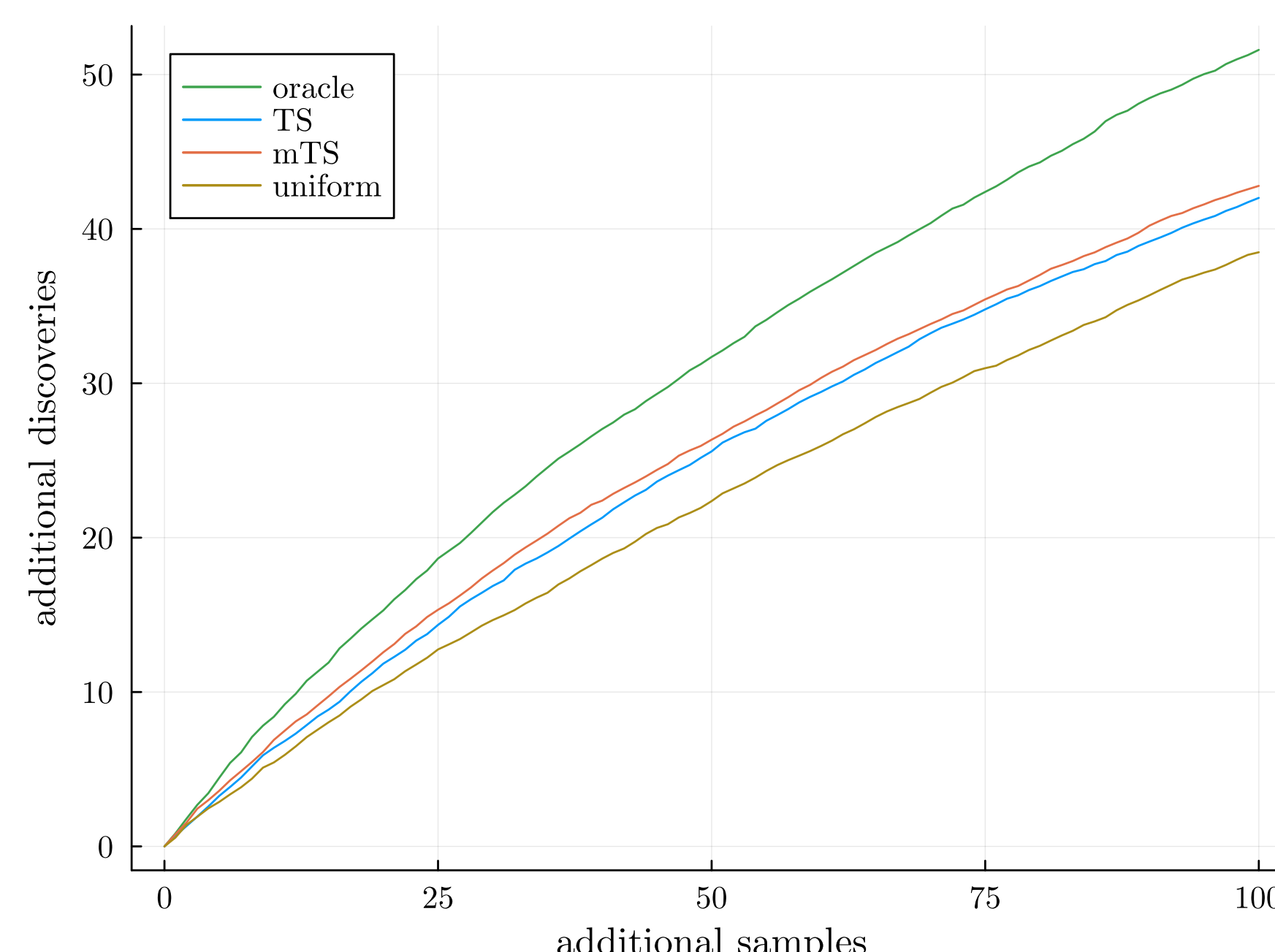
- total of 41 forests sites classified in $d = 4$ types, i.e. **groups** (EB, EC, DB, BC);
- around 100 **observations** per forest → trees counts in 10×10 m areas;
- records for 338 **species** across forest sites, of which 205 are **unique** to their forest type.

Preliminary analysis:

- cumulative number of **discovered** species;
- comparison with **oracle** and **uniform** strategies.

Sampling is initialized with 25 observations per forest type, during which 158 species are discovered on average.

Results are averaged over 100 experiments.



Conclusions

Simulation studies: sampling observations from the model,

- after **initial samples**, Thompson sampling strategies select populations with **greater biodiversity**;
- the TS and mTS strategies are often **nearly identical**.

Comparison with an **independent model**:

- for **high** traits sharing, the hierarchical model outperforms the independent model;
- for **low** traits sharing, they show similar performances → robust to model **misspecification**.

This non-standard setting is **overlooked** in the literature:

- lack of **competing** frequentist or Bayesian **methods**;
- lack of **theoretical properties** for optimal strategies → bounds on the **time-to-discovery**.

[1] Battiston, Favaro & Teh (2018). *Multi-Armed Bandit for Species Discovery: A Bayesian Nonparametric Approach*. Journal of the American Statistical Association.

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[4] Bubeck, Ernst & Garivier (2013). *Optimal discovery with probabilistic expert advice: Finite time analysis and macroscopic optimality*. Journal of Machine Learning Research.